

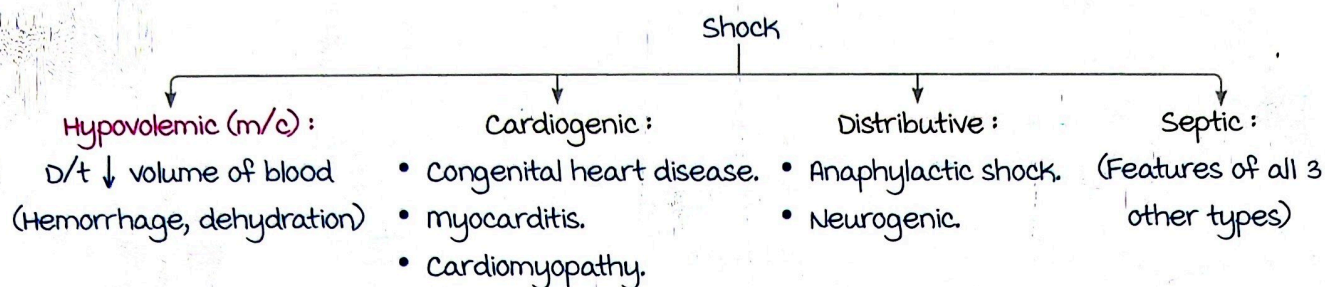
SHOCK

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Condition of circulatory dysfunction → Decreased oxygen delivery to tissue and decreased cellular perfusion insufficient for metabolic demands

Types of Shock

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STAGES OF SHOCK:

Compensated stage $\xrightarrow{\text{If prolonged}}$ Decompensated stage

- Vital organ function maintained (Brain, heart, kidney).
- mechanisms for BP maintenance:
Sympathetic activation → ↑ systemic vascular resistance (SVR) → ↑ contractility of heart
↓
Diverts blood from non-essential areas (Eg: Skin)
- Adequacy of circulation: Assessed by capillary refill time (CRT).
 - Done by blanching skin of thumb by pressure and noting refill time after.
 - Normal: ≤ 2 sec.
 - In shock: > 3 sec.

Compensated v/s decompensated shock:

Features	Compensated shock	Decompensated shock
mental status	Agitation or confusion	Drowsiness
Heart rate	Increased	markedly increased
CRT	Increased (> 3 sec)	Increased (> 5 sec)
urinary output	Normal	Oliguria/anuria
Blood pressure (BP)	Normal	Decreased
Respiratory rate	Increased	Increased

khanirfan0392@gmail.com

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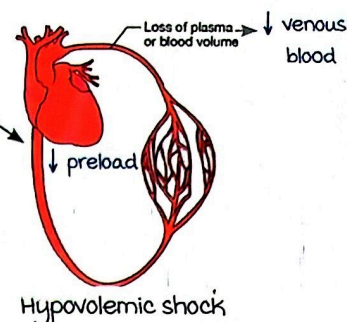
HYPOVOLEMIC SHOCK :

Suspected when compatible history of fluid loss / ↓ oral intake present.

Findings :

- Dehydration :
 - Dry mouth.
 - ↓ or absent tears.
 - Sunken eyes.
- ↓ renal perfusion :
 - Oliguria.
 - ↑ serum creatinine.
 - ↑ Blood urea.

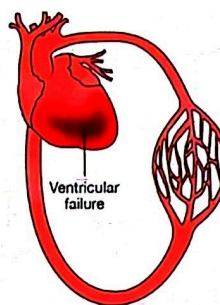
- ↓ central venous pressure (CVP).

**CARDIOGENIC SHOCK :**

Impaired cardiac contractility
↓
Venous pooling of blood

Findings :

- ↑ CVP.
- ↑ jugular venous pressure.
- S3 gallop.
- Congestive hepatomegaly.



Presentations of primary pathology of heart :

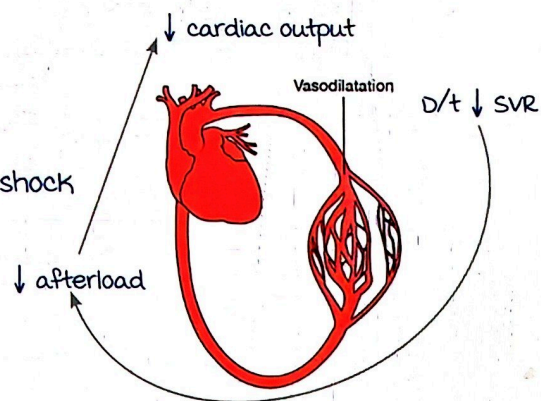
- murmurs.
- Friction rub (Pericarditis).

DISTRIBUTIVE SHOCK :

AKA warm shock.

Findings :

- Warm extremities.
 - Bounding pulses.
 - CRT - flash (< 1 sec).
- Called as warm shock

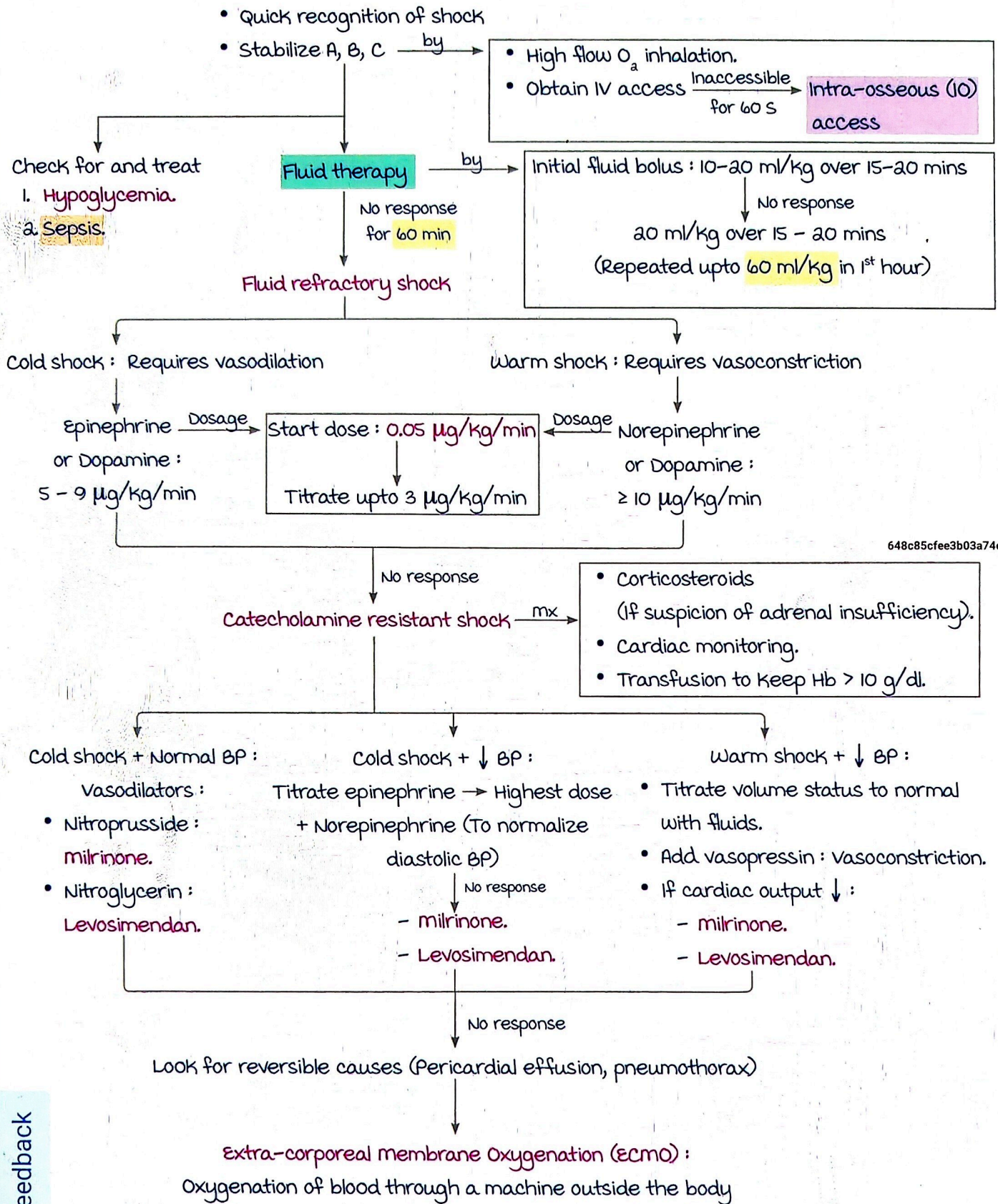
**SEPTIC SHOCK :**

	Cold type	Warm type
Extremities	Cold	Warm
CRT	CRT > 3 sec	CRT : Flash
Pulses	Feeble/decreased pulses	Bounding pulses

Management Algorithm for Shock

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Sites of IO access :

1. Proximal tibia.
2. Distal femur.

Rules of 60 :

- IO access to be obtained if IV **not available in 60 sec.**
- Fluid boluses upto 60 ml/kg in first 60 min.
- Start vasoactive drugs 60 min after starting fluid therapy.

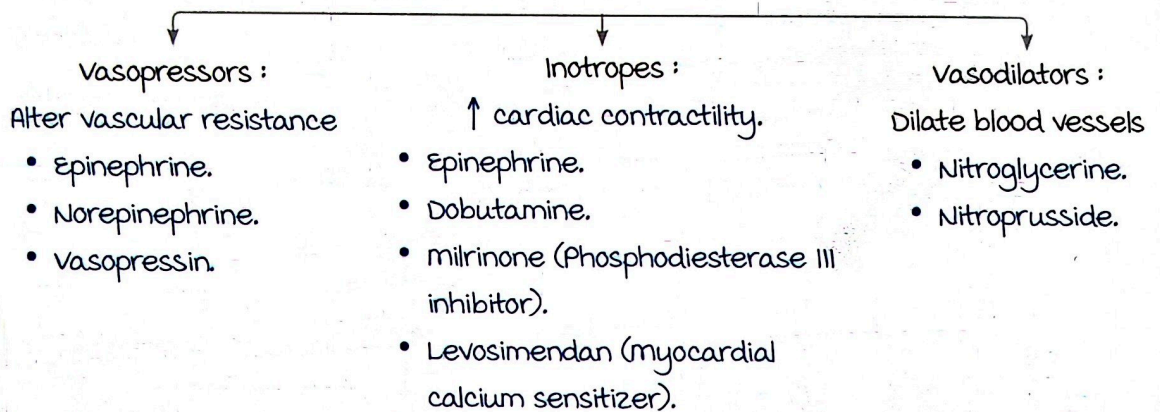
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Fluid therapy :

- Crystalloids (↓ price, easy availability) >> colloids (Long-term risk of usage in children : unknown).
 - Normal saline (0.9 %).
 - Ringer lactate.

Isotonic solutions only

management of sepsis : 3rd generation cephalosporin + vancomycin.

vasoactive drugs :**Therapeutic end-point :**

1. Normalization of vitals : Normal HR/BP.
2. urine output : > 1 ml/kg/hr.
3. CRT : < 2 sec.
4. mentation : Normal.